

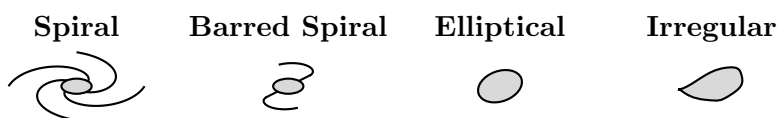
Machine Learning — Programming Assignment

July 2025

1 Problem definition and data

Understanding our universe is one of the most fascinating endeavors pursued by humankind. Thanks to modern telescopes, we can observe a multitude of galaxies, each one made of hundreds of billions of stars.

Galaxies are not all equal but come in a variety of shapes. Astronomers divide them into four categories:



A vast number of images of galaxies have been captured and manually classified. Here, we will use a dataset of 11 000 images equally divided in the four categories. Images have been converted to colors and resized to 224×224 pixels.

The dataset has already been divided into a training set of 10 000 images and a test set of 1000 images. The two sets have been organized in directories, with one subdirectory for each class.

2 Assignment

We want to build one or more classifiers that automatically identify the shape of a galaxy from its image.

For the programming assignment, you are expected to:

1. analyze and comment the data;
2. implement, train, and evaluate one or more classification models;
3. use suitable data processing and visualization techniques to analyze the behavior of the trained models.

All the above should be implemented as scripts in the Python programming language. The code written during the course can be used for the assignment if needed.

Special request: we are particularly interested in understanding how much color information can be useful for the classification.

Second special request: galaxies do not have a specific orientation and can appear rotated in any way. How can we exploit this fact to improve the classifiers?

3 Report

Prepare a report of three to five pages documenting all your work. Provide detailed instructions on how to reproduce the results. The report must be in the PDF format. Include your name in the report and conclude the document with the following statement: “I affirm that this report is the result of my own work and that I did not share any part of it with anyone else except the teacher.”

Make a ZIP archive with the report and the Python notebook (converted to PDF, with visible outputs), and submit it from the course web page.